MAVEN TOYS INTELLIGENCE REPORT

Maven Toys Intelligence Report

An end-to-end retail analytics project transforming raw sales and inventory data into actionable business insights using SQL, Python, and Power BI.

PHASE 1: SQL ANALYTICS

PHASE 2: PYTHON ADVANCED SCIENCE

PHASE 3: POWER BI FRAMEWORK

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Project Overview & Toolchain

An end-to-end retail analytics project analyzing sales performance, product performance, store performance, inventory health, and future sales trends to support data-driven decision-making.

SQL

Core data extraction & business analysis

Python

Pandas, NumPy, Matplotlib, Scikit-learn

Power BI & DAX

Executive dashboard framework & reporting

PHASE 1 · SQL ANALYTICS

Overall Sales Performance

$14.4M	1.09M	$4.0M
Total Revenue	**Units Sold**	**Net Profit**
Aggregate across all transactions	Total item volume moved	Revenue minus product cost

Calculated by joining sales against standard item margins: `Revenue = product_price × units`, `Profit = (price − cost) × units`.

PHASE 1 · SQL ANALYTICS

Top Products & Profitability

Product	Revenue
Lego Bricks	$2.39M
Colorbuds	$1.56M
Magic Sand	$969K
Action Figure	$927K
Rubik's Cube	$913K

Most Profitable Product

Colorbuds

$834,944 in profit — the single highest-margin product, fueling capital reinvestment.

Lego Bricks lead in revenue ($2.39M), but Colorbuds dominate profitability and volume with 104K units sold.

PHASE 1 · SQL ANALYTICS

Store Performance & Regional Dominance

Top Store

Ciudad de México 2 generates the highest revenue at **$554,553** — more than double the worst-performing store, Campeche 2 ($206K).

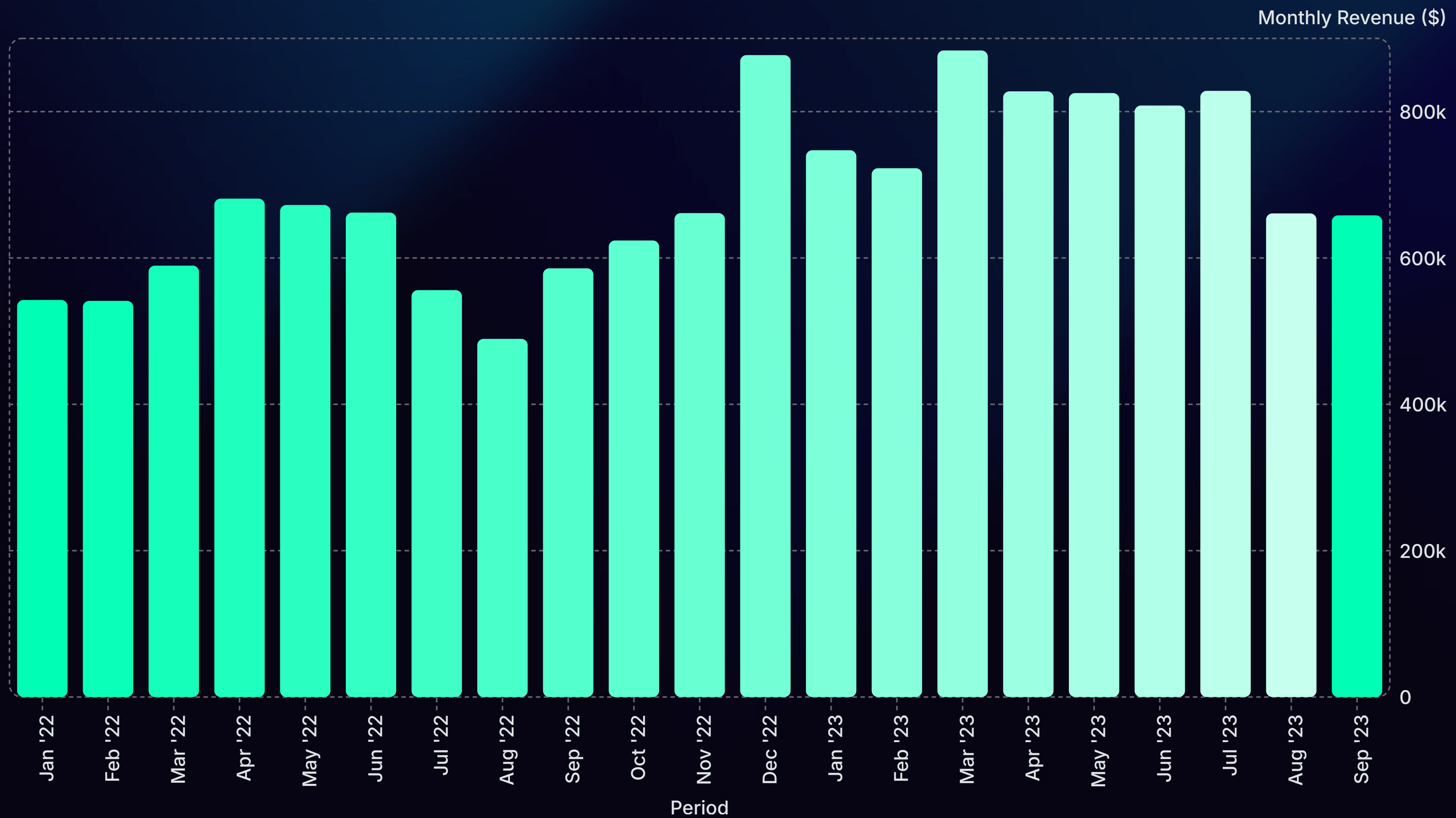
Top City

Ciudad de México dominates with **$1.65M** in total revenue, ranking as the top city by a significant margin.

Can the successful strategy deployed in Ciudad de México be replicated in mid-tier cities like Hermosillo and Guanajuato?

PHASE 1 · SQL ANALYTICS

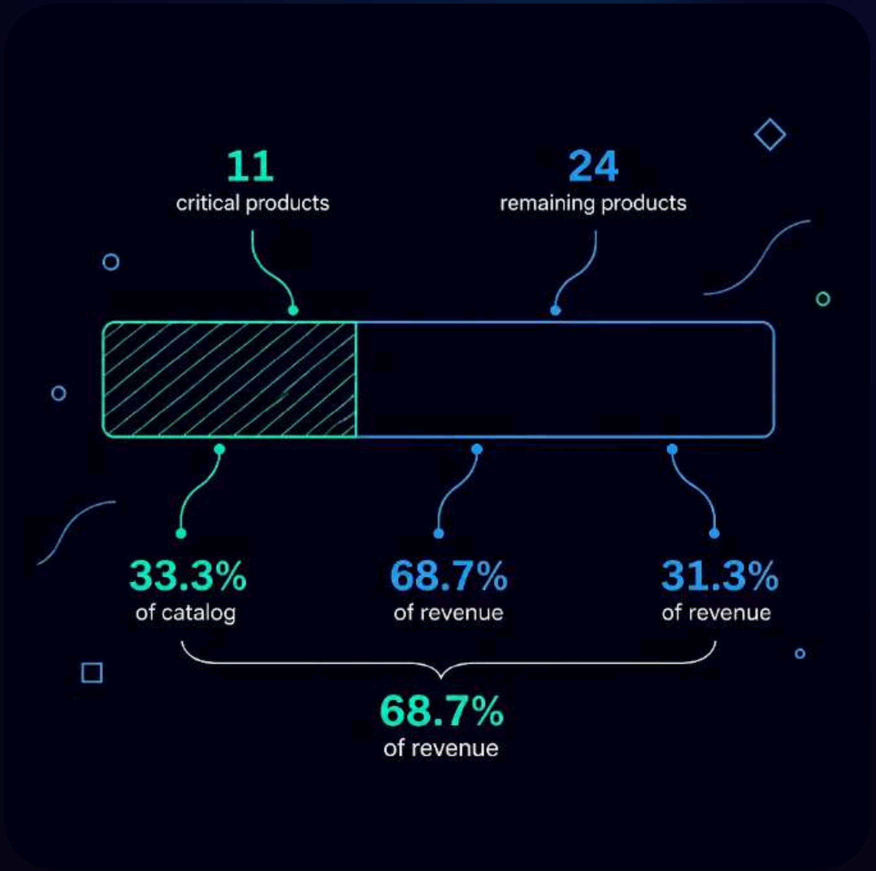
Seasonality & Cumulative Revenue



Cumulative revenue reached **$14.4M** by September 2023. A **sharp 20% drop in August 2023** ended the growth streak, stabilizing at ~$660K monthly.

PHASE 2 · PYTHON ANALYTICS

ABC Analysis: Pareto Concentration Risk



Critical Finding

Just **11 high-velocity products** — less than a third of the catalog — capture **68.7% of total revenue**.

Operational Mandate

Transition to hyper-focused replenishment for these SKUs. Stockouts here cause severe top-line damage; lower-tier items have negligible bottom-line impact.

PHASE 2 · PYTHON ANALYTICS

Stockout Risks & Store Clustering

20 Products at Immediate Risk

Zero inventory remaining with positive daily demand — triggering unrecoverable lost sales and degraded customer loyalty.

Cluster 2: High Velocity

Mean revenue $444K, 33.8K volume. Priority replenishment mandatory.

Cluster 1: Mid-Market

Mean revenue $282K, 21.2K volume. Optimize product mix to increase turns.

Cluster 0: Lean Volume

Mean revenue $265K, 20K volume. Monitor slow-moving inventory closely.

One overstocked product hoards 2,200+ units with subpar sales — a prime candidate for aggressive markdowns.

PHASE 3 · POWER BI DASHBOARD

Executive Insights & Strategic Gaps

Revenue Categories

Toys dominate at $5.09M. Electronics ($2.25M) and Games ($2.23M) trail closely as secondary drivers.

Inventory Imbalance

Art & Crafts holds 29% of stock; Electronics only 8%. Average of just 15 days of inventory remaining.

Inter-Store Transfers

Ciudad de México 2 holds 915 units while Aguascalientes 1 and Campeche 1 sit at 0 days remaining.

Stagnant Forecast

Month-over-month growth at −0.41%. 3-month forecast projects flat revenue with no upward trend.

Key Takeaways & Next Steps

Protect SKUs

Hyper-focus replenishment on 11 critical SKUs

Resolve Stockouts

Fix 20 immediate stockouts with auto POs

Enable Transfers

Implement inter-store stock transfers

Activate Campaigns

Launch promotions to break flat forecast

The data is clear: protect what drives revenue, fix what's bleeding, and innovate to reignite growth.